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The Trend on The Bag's Production

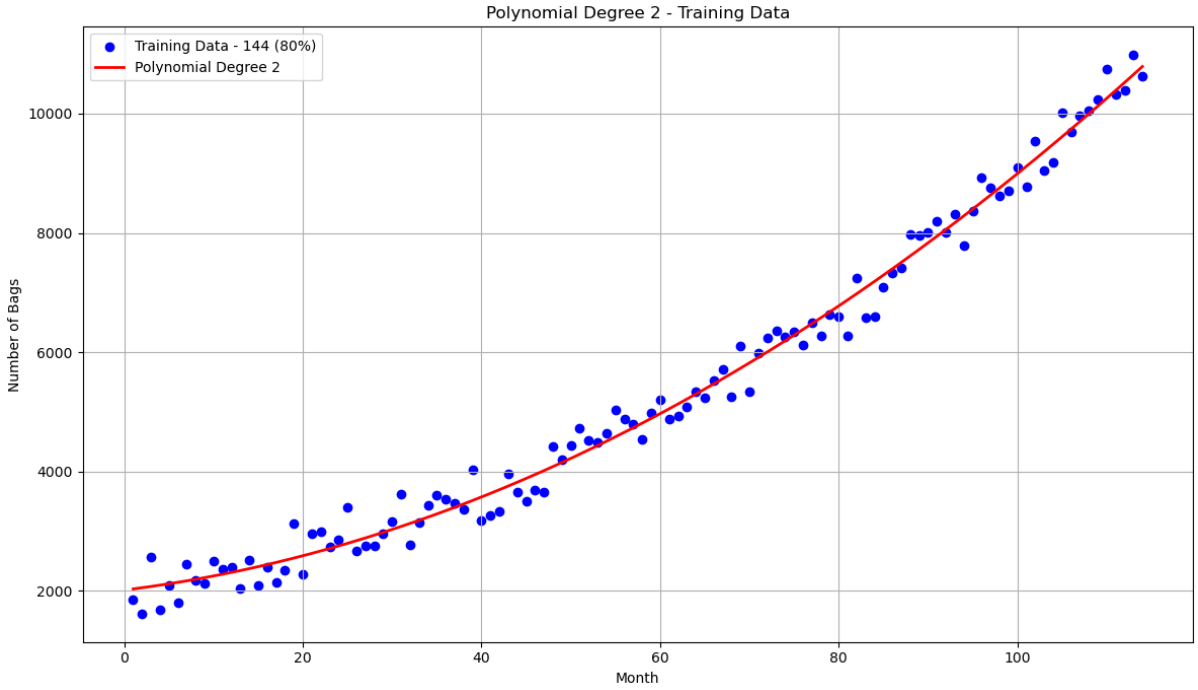
I compared two methods to evaluate their accuracy: polynomial of degree 2 and polynomial of degree 3:

To find the mathematical model equation, the data needs to be divided into training and testing sets. I used 80% of the data, which is 114 data points, for training, and 20%, which is 30 data points, for testing. This approach is intended to prevent overfitting.

I used the scikit-learn library to build the mathematical models, specifically the PolynomialFeatures and LinearRegression classes. PolynomialFeatures was used to generate the polynomial terms, which were then fed into LinearRegression.

1. Polynomial of degree 2
   The reason I choose this method because the data is not linear, but rather forms a single curve (like a parabola opening upward or downward). This shape is ideal for representing trends that have a single peak or trough.
   Mathematical models: $y = ax^2 + bx + c$
   a = 0,5130
   b = 18,4852
   c = 2014,8884
   $y = 0{,}5130x^2 + 18{,}4852x + 2014{,}8884$

   Note:
   y = predicted production quantity
   x = time/monthly sequence

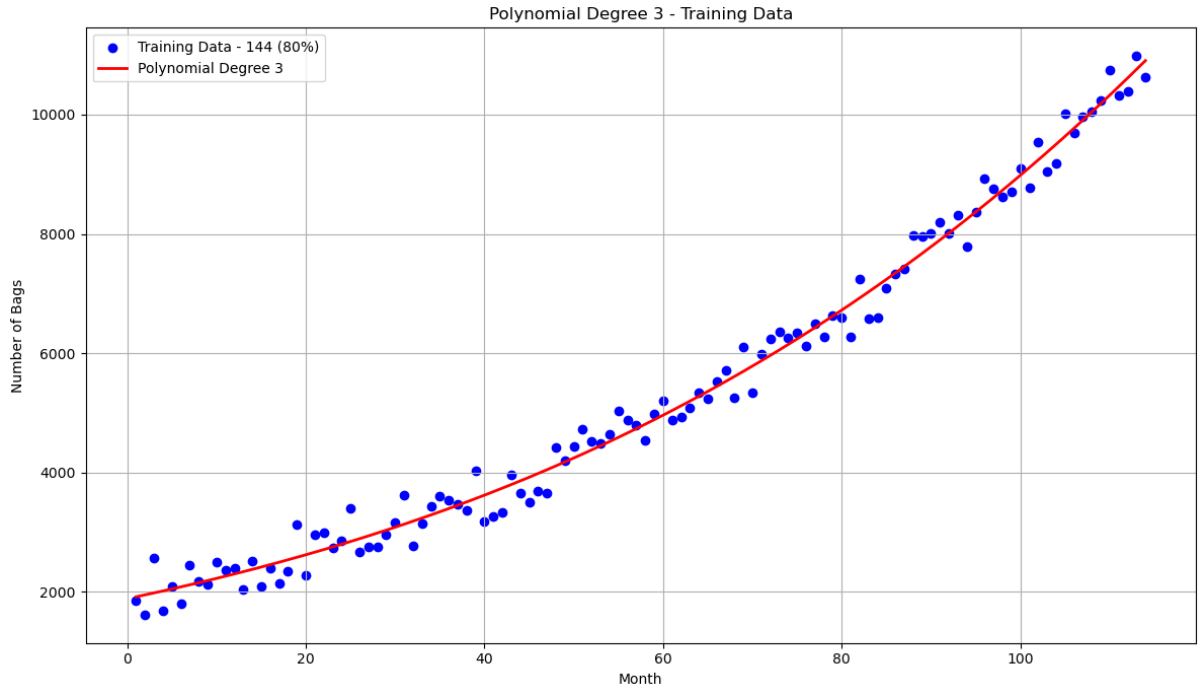


Mean Absolute Error (MAE) : 230,9165
Mean Squared Error (MSE) : 77.047.7911

Root Mean Squared Error (RMSE) : 277.5748
R-squared (R2) : 0,9887

2. Polynomial degree 3
   The reason I choose this method because the data exhibits a more wavy pattern or an 'S'-shaped curve, indicating that it has more than one bend. This allows it to capture changes in the rate of a trend, such as growth that starts slowly, then accelerates rapidly, and eventually levels off.
   Mathematical models: $y = ax^3 + bx^2 + cx + d$
   a = 0,0016
   b = 0,2298
   c = 31,5678
   d = 1886.7842
   $y = 0{,}0016x^3 + 0{,}2298x^2 + 31{,}5678x + 1886.7842$

   Note:
   y = predicted production quantity
   x = time/monthly sequence



   Mean Absolute Error (MAE) : 228,4672
   Mean Squared Error (MSE) : 74.937,6761
   Root Mean Squared Error (RMSE) : 273,7475
   R-squared (R2) : 0,9890

Comparison of evaluation metrics:

a. Mean Absolute Error (MAE)
   Degree 2: 230,9165
   Degree 3: 228,4672
   Model Level 3 has a lower MAE value, indicating that its predictions deviate by an average of approximately 228,47 points from the actual values. In contrast, Model Level 2 has an average

deviation of about 230,92 points. Model Level 3 demonstrates slightly better performance due to its lower average error.

b. Mean Squared Error (MSE)
   Degree 2: 77.047,7911
   Degree 3: 74.937,6761
   Model Level 3 has a lower MSE value, indicating that, in addition to having a smaller average error, it is also more consistent and less likely to produce large errors compared to Model Level 2. This suggests that Model Level 3 is more effective in minimizing extreme errors.

c. Root Mean Squared Error (RMSE)
   Degree 2: 277,5748
   Degree 3: 273,7475
   The RMSE value of Model Level 3 is lower, indicating that its typical prediction error is around 273,75 points, which is smaller than that of Model Level 2 at 277,57 points. Therefore, Model Level 3 is considered better as it generally produces smaller prediction errors.

d. R-squared (R2)
   Degree 2: 0,9887
   Degree 3: 0,9890
   Model Level 3 has a slightly higher R^2 value, being able to explain 98,90% of the variation in the data, compared to Model Level 2 which explains only 98,87%. This means that Model Level 3 has a slightly better ability to represent the patterns present in the data.

In conclusion, the polynomial of degree 3 performs better than the polynomial of degree 2.